A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Ozone depletion - Set 1

Mark scheme

AQA spec 3.2.1.3; The physical environment; textbook pp. 108-113

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 3 points below. **1** UV light splits diatomic oxygen (O2) into two monatomic oxygen atoms (O) **1** Monatomic oxygen reacts with diatomic oxygen to form ozone (O3) **1** UV light also splits ozone into O2 and O – the reactions form a dynamic equilibrium which absorbs UV (converting it to chemical energy / heat) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
01.2	**Marks shared** Up to 6 marks in total - share them between the 4 points below according to how completely each one is made. **up to 2 Useful:** boiling points near ambient temperature so easily liquefied under low pressure (aerosol propellants, fridge / air-conditioning coolants) – dissolve grease but do not damage electrical components (solvents) – non-flammable – non-toxic **up to 2 Persistence:** chemically stable / unreactive in the troposphere so they remain long enough to be carried up to the stratosphere **up to 2 Dissociation by UV:** in the stratosphere higher UV levels break the carbon-chlorine bonds releasing chlorine free radicals **up to 2** Chlorine reacts with monatomic oxygen (and ozone), preventing ozone formation – chlorine is regenerated so one atom destroys many ozone molecules – bromine / iodine act similarly **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **CFCs** → CFCs / halogenated organic compounds releasing chlorine • **Persistence** → does not break down / stays in the environment for a long time **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	6
01.3	**1 mark each** 1 mark for each of the 3 points below. **1** UVB energy is absorbed by cells and breaks up biological molecules / DNA causing mutations, skin cancer, cataracts, sunburn (human health) **1** Damage to leaf tissue and reduced photosynthesis / crop yields (plants) **1** Damage to marine algae, phytoplankton, corals and planktonic larvae (marine organisms), reducing productivity and food webs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
01.4	1 mark for the point below. **1** An area where the total ozone column falls below 220 Dobson Units (normal is about 300 DU) - a level never recorded before CFC pollution **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1

Do not write outside the box

Question	Marking guidance	Marks
01.5	1 mark for the point below. **1** Concentration varies with altitude and the stratosphere is not normally sampled – Dobson Units measure the total column thickness instead **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	1
01.6	1 mark for the point below. **1** The British Antarctic Survey at Halley detected higher UV at ground level, showing ozone above had decreased (satellites above the stratosphere cannot measure downward UV directly) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-109; AQA spec 3.2.1.3.	
02.1	**1 mark each** 1 mark for each of the 2 points below. **1** Minimum ozone fell steeply from about 225 DU in 1979 to about 95 DU in the mid 1990s **1** Since then it has fluctuated around 100-115 DU with a slight recovery / no further decline **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (225 - 95) / 225 x 100 **1** = 57.8% (accept 58%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
02.3	**1 mark each** 1 mark for each of the 4 points below. **1** Stratospheric temperatures over Antarctica are the lowest on Earth so stratospheric clouds / ice crystals form **1** Ice crystal surfaces allow reactions that convert the chlorine released from CFCs into chlorine molecules (Cl_2) during the dark winter **1** The polar vortex (rotating winds) prevents mixing with the rest of the atmosphere, keeping temperatures low and chlorine concentrated **1** When sunlight returns in spring (September) it splits the chlorine molecules into free radicals that destroy ozone – levels recover by December as the vortex breaks down **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **CFCs** → CFCs / halogenated organic compounds releasing chlorine **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	4
02.4	**1 mark each** 1 mark for each of the 2 points below. **1** Ground and satellite UV / ozone measurements show depletion but not the chemistry / correlation is not causation **1** Air samples collected from the stratosphere by helium balloons and high-flying research aircraft detected chlorine and chlorine monoxide, confirming the Rowland-Molina hypothesis **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **CFCs** → CFCs / halogenated organic compounds releasing chlorine • **correlation** → a relationship in which the two variables change together • **causation** → one variable actually causing the change in the other **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated • a correlation presented as proof of cause - say that it suggests a link that further work would have to test	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 110-111; AQA spec 3.2.1.3.	

Do not write outside the box

Question	Marking guidance	Marks
03.1	**1 mark each** 1 mark for each of the 3 points below. **1** Manufacture and use of CFCs and other ozone-depleting substances (ODSs) phased out then banned **1** HCFCs to be phased out (by 2030) – some essential uses (eg halon fire extinguishers in aircraft) still permitted **1** A fund to help (developing) countries implement the Protocol – legally binding targets (unlike the 1985 Vienna Convention) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **CFCs** → CFCs / halogenated organic compounds releasing chlorine **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	3
03.2	**2 marks per point** 2 marks for each of the 2 points below; award 1 mark where a point is only partly made. **2 Aerosols:** pump-action sprays / roll-on deodorants (alternative process), or hydrocarbon propellants (propane, butane) - flammable / fire risk **2 Refrigerants:** HCFCs (less stable so break down in the troposphere, but still some ozone depletion) then HFCs (no chlorine) - more expensive, less effective, powerful greenhouse gases **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	4
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** CFCs are drained from the appliance and incinerated / broken down at high temperature **1** Products are CO2 and acidic gases (HCl, HF) which are neutralised with crushed lime **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **CFCs** → CFCs / halogenated organic compounds releasing chlorine **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	2

Do not write outside the box

Question	Marking guidance	Marks
03.4	1 mark for the point below. **1** CFCs are persistent, so those already released continue to reach the stratosphere and destroy ozone for many years – some ODSs (eg dichloromethane) are not banned **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ozone layer** → the ozone in the stratosphere that absorbs UV • **CFCs** → CFCs / halogenated organic compounds releasing chlorine • **persistent** → does not break down / stays in the environment for a long time **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 112-113; AQA spec 3.2.1.3.	

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Question	Marking guidance	Marks
04.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • Stages of scientific method illustrated by ozone: identification (Antarctic UV data), hypothesis (Rowland-Molina 1974), data collection / analysis (Dobson units, satellites, balloons confirming ClO), proposed solutions, enactment (Vienna 1985, Montreal 1987) • **Evidence of success:** ozone hole area and depth stabilised and beginning to shrink – CFC concentrations in the atmosphere falling – UV levels not rising further – global ozone loss limited to about 4% • **Reasons for success:** international recognition of serious consequences (skin cancer) – agreement of nearly every country – legally binding targets with a fund for developing countries – relatively few producers / products – alternatives (HCFCs, HFCs, pump sprays, hydrocarbons) already available and cheap – no lifestyle change needed • **Limits:** persistence of CFCs means recovery takes decades – some ODSs still used – HFC replacements are greenhouse gases • **Climate change contrast:** greenhouse gases come from almost every human activity (energy, transport, agriculture, industry) in every country – fossil fuels underpin economies so control requires major lifestyle and economic change • Agreements (Kyoto, Paris) have weaker / non-binding targets, disputes between developed and developing nations – emissions still rising – natural fluctuations, time delays and tipping points make cause-effect less clear to the public	9

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Question	Marking guidance	Marks
	• **Judgement:** ozone control was a technical substitution problem – climate change is a systemic one, so the ozone success is only partly transferable **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • The converse of any point, ie the same comparison written the other way round. • **ozone layer** → the ozone in the stratosphere that absorbs UV • **persistence** → does not break down / stays in the environment for a long time • **CFCs** → CFCs / halogenated organic compounds releasing chlorine **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	
	Model answer - Level 3 (9 marks)	
	Ozone depletion followed the scientific method almost textbook-fashion: Antarctic UV and ozone measurements identified a problem, Rowland and Molina's 1974 hypothesis explained it, Dobson-unit, satellite and balloon data confirmed chlorine monoxide in the stratosphere, and the Montreal Protocol enacted a solution. It has largely worked: CFC concentrations are falling, the ozone hole has stabilised and is starting to shrink, and global ozone loss was limited to about 4%.	
	Success came from particular conditions. The consequence - skin cancer - was clear and frightening, nearly every country agreed, targets were legally binding with a fund to help developing countries, there were relatively few producers and products, and alternatives (HCFCs, HFCs, hydrocarbons, pump sprays) already existed, so substitution cost little. Even so recovery will take decades because CFCs are so persistent, some ozone-depleting substances are still in use, and the HFC replacements are themselves greenhouse gases.	
	Climate change has none of these advantages. Greenhouse gases come from almost every activity - energy, transport, agriculture, industry - in every country, and fossil fuels underpin entire economies, so control means changing lifestyles and economies rather than swapping one chemical for another. Kyoto and Paris set weaker, largely non-binding targets, developed and developing nations dispute responsibility, and natural fluctuations, time lags and tipping points make cause and effect less obvious to the public. Emissions are still rising. The ozone story shows international agreement can work, but it was a technical substitution problem; climate change is systemic, so its lessons transfer only partly.	
	Weaker answer - Level 2 (4 marks)	
	Ozone depletion was caused by CFCs. Scientists found the hole over Antarctica and the Montreal Protocol banned CFCs, so the ozone layer is now recovering. This was successful because everyone agreed and there were alternatives to CFCs. Climate change has not been tackled as well. There have been agreements like Kyoto and Paris but carbon dioxide is still going up. This is because fossil fuels are used everywhere and countries do not agree. So ozone depletion was solved better than climate change.	

Do not write outside the box

Question	Marking guidance	Marks
	What makes the difference	
	• Level 2 gives the right outline but Level 3 supports it with evidence (falling CFC levels, 4% loss, Dobson units, ClO detection) and explains each reason for the difference.	
	• Level 3 acknowledges the limits of the ozone success (persistence, HFCs) - balance is expected at the top level.	
	• The final judgement in Level 3 characterises the difference (substitution vs systemic problem); Level 2 just restates the question.	
04.2	1 mark for the point below. **1** Vienna Convention for the Protection of the Ozone Layer - no legally binding goals **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Ozone Layer** → the ozone in the stratosphere that absorbs UV **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-113, 105-106; AQA spec 3.2.1.3, 3.2.1.2.	
05.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** C **Also accept** • **stratospheric ozone** → the ozone in the stratosphere that absorbs UV	1
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** Stratospheric ozone is natural and beneficial: absorbs UV, protecting living organisms **1** Tropospheric ozone is a secondary pollutant (from NOx photochemistry) that is toxic to plants and humans and is a greenhouse gas **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **Stratospheric ozone** → the ozone in the stratosphere that absorbs UV	2

Do not write outside the box

Question	Marking guidance	Marks
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** HCFCs are less chemically stable / contain hydrogen so they are broken down in the troposphere **1** So most do not persist long enough to reach the stratosphere where UV would release their chlorine **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **CFCs** → CFCs / halogenated organic compounds releasing chlorine **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-112; AQA spec 3.2.1.3.	

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